

eCOA Solutions

Product Requirements Document (PRD)

Electronic Clinical Outcome Assessment Platform

Document Version	1.0
Date	December 8, 2025
Status	Draft
Classification	Internal

1. Executive Summary

This Product Requirements Document outlines the specifications for the eCOA (Electronic Clinical Outcome Assessment) Solutions platform. The platform enables pharmaceutical companies and clinical research organizations to create, deploy, and manage electronic forms for capturing patient-reported outcomes, caregiver assessments, and investigator evaluations during clinical trials.

The solution focuses on providing a sponsor-centric experience with robust template management, multi-language support, notification services, and compliance with pharmaceutical regulatory requirements including digital workflow instructions (DWI) for pharma manufacturing.

2. Product Vision

2.1 Problem Statement

Clinical trials require systematic collection of patient outcomes through standardized questionnaires. Current solutions lack flexibility in template adoption, have limited multi-language support, and do not provide adequate sponsor control over the study lifecycle from UAT to production deployment.

2.2 Product Goals

- Enable sponsors to create, manage, and deploy clinical studies with ease
- Provide flexible template adoption from licensed questionnaires or custom templates
- Support multi-language forms for global clinical trials
- Ensure legal compliance with pharmaceutical manufacturing regulations
- Deliver robust notification and reminder services for patient engagement

3. User Personas

3.1 Sponsor/Study Administrator

Primary user responsible for creating studies, managing templates, deploying forms to production, and overseeing data collection. Has access to sponsor login, study dashboard, and administrative functions.

3.2 Patient

End user who completes electronic forms on mobile or web applications. Receives notifications and reminders for form completion. Identified by unique patient code within each study.

3.3 Caregiver

Assists patients in completing forms or completes caregiver-specific assessments. May receive delegated access to patient forms.

3.4 Investigator

Clinical site personnel who complete investigator-rated assessments and may review patient submissions.

4. Functional Requirements

4.1 Authentication & Authorization

4.1.1 Sponsor Login

1. System shall provide secure login with username and password authentication
2. Upon successful login, sponsor shall see a list of all their studies (Study 1, Study 2, Study 3, etc.)
3. System shall provide "Create New" button to initiate new study creation
4. Role-based access control for sponsor, data management, and trial manager roles

4.2 Study Management

4.2.1 Create New Study

When creating a new study, the system shall present two options:

- **Use Templates:** Select from existing templates
- **Create from Scratch:** Build forms using blank canvas with drag-and-drop tools

4.2.2 Template Options

The template selection screen shall provide:

1. **Clone Study:** Option to clone an already deployed or draft study
2. **Licensed Questionnaires:** Select from list of licensed/validated questionnaires
3. **Company Templates:** Pre-defined templates specific to the organization (e.g., Cipla Templates)
4. **Custom Templates:** Option to create and save own templates for reuse

4.2.3 Template Search & Clone

- Search functionality to find questionnaires/studies by name
- Display results showing Quest name, Study association, and Clone action
- After cloning, navigate to Form Properties for configuration

4.3 Form Builder

4.3.1 Canvas & Tools Panel

The form builder shall provide a blank canvas with a Tools panel containing:

- Radio buttons (single select)
- Checkboxes (multi-select)
- Scale 1, Scale 2, Scale 3 (rating scales)
- Text input fields

4.3.2 Properties Panel

Each form element shall have configurable properties:

- Font selection
- Size adjustment
- Color customization
- Alignment (Left, Center, Right)
- Option to add logos

4.3.3 Page Navigation

1. Add "Next Page" button on all pages
2. Add "Back" button on every page

3. Drag-and-drop interface for adding questions
4. **Mandatory Review Page:** Last page of all forms must display review of all answers

4.4 Form Properties Configuration

4.4.1 Completion Settings

Property	Options
Completed By	Patient, Caregiver, Investigator
Notification Frequency	Twice a day, Select specific time & frequency
Include in Reports	Yes / No
Schedule	Select schedule with different tabs/views

4.4.2 Form Triggers

Forms shall support various trigger frequencies:

- Daily triggers
- Weekly triggers
- Monthly triggers
- Custom notification signals and alerts

4.5 Deployment Workflow

4.5.1 UAT to Production Pipeline

1. Sponsor completes form/study configuration
2. Study enters UAT stage for sponsor validation
3. Priority-based study deployment (drag-and-drop ordering)
4. Change request management with title and tracking
5. Production deployment
6. Patient data collection begins with unique patient codes

4.5.2 Study Deployment Capacity

- Support multiple studies per sponsor (Study 1, Study 2, Study 3...)
- App store deployment integration
- Production monitoring and management

4.6 Patient Application

4.6.1 Mobile & Web Access

- Forms accessible via mobile application
- Browser-based access option
- Unique patient code for form access and identification
- Dynamic form rendering (similar to Google Forms experience)

4.6.2 Notification & Reminder Service

- Push notifications for form completion reminders
- Configurable reminder frequency and timing
- Alarm selection based on form schedule

4.7 Language Support

4.7.1 Multi-Language Forms (Phase 2)

- Support regional languages within forms (e.g., Marathi for Maharashtra hospitals)
- Same form, same questions published globally in multiple languages
- Language support within form properties configuration

4.8 Data Management & Reporting

4.8.1 Data Storage & Retrieval

- Data storage with sponsor database access (primary and secondary access keys)
- Minimal data storage during deployment phase
- Data retrieval and validation mechanisms
- Global data validation support

4.8.2 Reporting Functionality

- Standard reporting capabilities
- Project dashboard for study overview
- Include/exclude forms from reports via form properties

5. Non-Functional Requirements

5.1 Compliance & Regulatory

- Legal compliance with pharmaceutical manufacturing regulations
- Digital Workflow Instruction (DWI) support for pharma manufacturing
- 21 CFR Part 11 compliance for electronic records
- Audit trail for all form submissions and changes

5.2 Security

- Secure sponsor authentication
- Unique patient codes for data protection
- Database access management with primary/secondary keys
- HTML-level backend validation

5.3 Performance

- Minimal data storage for efficient performance
- Support for multiple concurrent studies
- Responsive form rendering on mobile devices

6. MVP Scope (Phase 1)

The Minimum Viable Product focuses on core functionality:

1. Sponsor login and study list management
2. Basic form builder with essential tools (Radio, Checkbox, Scale, Text)
3. Template adoption and cloning functionality
4. Dynamic forms with unique patient codes
5. Basic notification service
6. UAT to Production deployment workflow
7. Standard reporting functionality
8. Single language support

7. Phase 2 Enhancements

1. Multi-language form support
2. Advanced vendor/platform integrations
3. Enhanced change request management
4. Advanced analytics and dashboards
5. API integrations with external systems

8. User Flow Summary

8.1 Sponsor Study Creation Flow

1. Sponsor logs in with username/password
2. Views list of existing studies
3. Clicks "Create New" to start new study
4. Selects "Use Templates" or "Create from Scratch"
5. If template: Searches and clones existing questionnaire/study
6. Configures form properties (completion by, notifications, schedule)
7. Finalizes form design with review page
8. Deploys to UAT for testing
9. Promotes to Production
10. Monitors patient data collection

9. Appendix

9.1 Glossary

Term	Definition
eCOA	Electronic Clinical Outcome Assessment
UAT	User Acceptance Testing
DWI	Digital Workflow Instructions
MVP	Minimum Viable Product
Sponsor	Pharmaceutical company or CRO managing the clinical trial

9.2 Document History

Version	Date	Author	Changes
1.0	Dec 8, 2025	Product Team	Initial draft based on design meeting